NovoTech MACHINE TOOLS Inc.

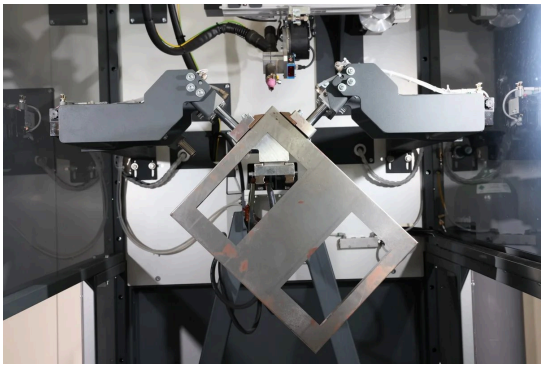
PRODUCT CATALOGUE 2026

GECAM · MADE IN ITALY · SOLD & SERVICED IN CANADA BY NOVOTECH

CORNER WELDING MACHINES

GECAM 1-, 2-, 3- AND 4-AXIS TIG AND PLASMA CORNER WELDERS FOR BENT BOXES — CWS 200 · CWS 300 · CWS 300 XY · WELCOR 400 X · XY · XYZ · XYZ-Φ







Italy
MADE IN ITALY

4
MACHINE FAMILIES

1-year
WARRANTY

Canada
PARTS & TECHNICAL SUPPORT

GECAM CORNER WELDING MACHINES

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Specifications subject to change without notice. Imperial values are converted from the manufacturer's metric data and rounded: thin sheet in gauge (ga), plate and capacities in common fractions (1/2, 1/4, 1/8, 1/16); metric values in brackets are the manufacturer's nominal figures. Model numbers are manufacturer designations. Photos may show optional equipment.

ABOUT NOVOTECH

YOUR CANADIAN SOURCE FOR GECAM CORNER WELDING MACHINES

Sales, installation, parts and technical support across Canada.

NovoTech Machine Tools Inc. supplies metalworking and fabrication machinery to job shops, fabricators, maintenance departments and manufacturers across Canada from our base in Mississauga, Ontario, Canada. Every machine in this catalogue is sold and supported locally by NovoTech — with spare parts and technical support readily available in Canada.

INCLUDED WITH EVERY NOVOTECH MACHINE

- Made in Italy by GECAM — 100 % Italian
- CSA-approved electrical panel
- 1-year warranty
- 7-day money-back guarantee
  in-stock machines only
- Spare parts & technical support
  readily available in Canada
- Order today with just 10% down
  0% down financing available
- Installation and training available

GECAM · PARMA, ITALY

CORNER WELDING, MADE IN ITALY

GECAM has designed and built machines for deburring, grinding and finishing metal at Vicofertile, Parma, since 1994 — 100 % made in Italy, with components from Italian suppliers within 125 miles of the plant, and CE-conforming to the Machinery, EMC and Low Voltage directives. Its corner welders weld the corners of bent boxes, cabinets, trays and enclosures with the torch driven by the machine: from the one-axis CWS for a straight corner seam to the four-axis WELCOR that reads the seam with a laser before it welds.

NovoTech sells, installs and supports GECAM corner welders in Canada from Mississauga, Ontario — copper groups and the welding power source chosen for your parts, installation, operator training, a CSA-approved panel, a 1-year warranty and parts and technical support in Canada.

GECAM CORNER WELDERS · 1 TO 4 AXES

WHICH CORNER WELDER? EVERY MODEL SIDE BY SIDE

The number of axes decides what kind of corner the machine welds in one cycle: one axis a straight seam, two and three axes a seam that runs in two or three directions, four axes adds torch rotation and a laser that reads the seam. Every model welds mild steel, stainless steel and aluminum with a TIG or plasma source at 0 – 47 in/min (0 – 20 mm/s), the part clamped pneumatically from inside against copper. Each section has its own pages further on, with the full specification table.

Model	Axes	Max. corner height	Part size	Control	Best for
CWS 200	1 — X	7-7/8" (200 mm)	3-15/16" – 59-1/16" square	Touch panel, recipes	Plain boxes, corners to 7-7/8"
CWS 300	1 — X	11-13/16" (300 mm)	3-15/16" – 59-1/16" square	Touch panel, recipes	Plain boxes, corners to 11-13/16"
WELCOR 400 X	1 — X	15-3/4" (400 mm)	3-15/16" – 39-3/8" square	Numerical control, 10 parameters	Deeper boxes, corners to 15-3/4"
CWS 300 XY	2 — X, Y	7-7/8" – 11-13/16" (200 – 300 mm)	3-15/16" – 59-1/16" square	Touch panel: X, Y positions, recipes	Large parts, torch driven on X and Y
WELCOR 400 XY	2 — X, Y	X 15-3/4" · Y 4-5/16" (400 · 110 mm)	3-15/16" – 39-3/8" square	Numerical control, 10 parameters	Seams in two directions
WELCOR 400 XYZ	3 — X, Y, Z	X 15-3/4" · Y 4-5/16" · Z 1-15/16" (400 · 110 · 50 mm)	3-15/16" – 39-3/8" square	Numerical control, 10 parameters	Seams in three directions
WELCOR 400 XYZ-Φ	4 — X, Y, Z, torch rotation	X 15-3/4" · Y 4-5/16" · Z 1-15/16" (400 · 110 · 50 mm)	3-15/16" – 39-3/8" square	Color touch screen, laser seam reading	Repeat series, self-learning seam path

Copper groups The corner is held square and kept cool by copper: on the CWS and the WELCOR 400 to three axes an outer unit 7-7/8" or 11-13/16" (200 or 300 mm) high and an inner unit of 1-15/16", 2-15/16", 3-15/16", 4-15/16", 5-7/8", 7-7/8" or 11-13/16" (50 – 300 mm), cooled or not; on the four-axis WELCOR modular aluminum and copper pads 3/8" to 7-7/8" (10 – 200 mm) thick. Cooled inner groups are recommended to avoid dark marks on the parts.

Corner preparation Bend the corner so the two edges meet on it — .0004" (0.01 mm) apart and centred on the corner — or, where the bend does not allow it, so they overlap by half the sheet thickness at most. On the CWS the gap between the edges must not exceed .008" (0.2 mm).

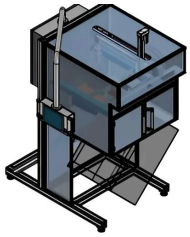
Welding power source Not included. GECAM offers the Lorch T 250 TIG source — T 250 DC for mild and stainless steel, T 250 AC/DC also for aluminum — with a cooling unit and a torch group with a 9' 10" (3 m) cable; the machines also work with a plasma source.



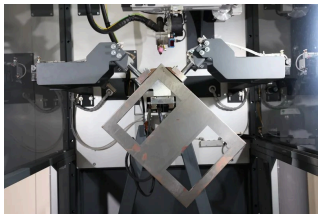
GECAM CWS 300 one-axis corner welding machine



GECAM CWS 300 XY two-axis corner welding machine



GECAM WELCOR 400 corner welder



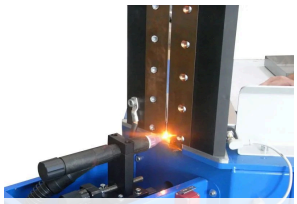
GECAM WELCOR 400 4-axis welding the corner of a square frame

CWS 200 · CWS 300 · WELCOR 400 X · 3 MODELS

CWS 200 · CWS 300 · WELCOR 400 X 1-AXIS CORNER WELDERS, CORNERS TO 15-3/4" HIGH



MODEL 1-AXIS · GECAM CWS 300 ONE-AXIS CORNER WELDING MACHINE



TIG torch welding a corner along the copper guide



Welded corner

One axis · corners 7-7/8", 11-13/16" or 15-3/4" high · parts to 59" square · TIG or plasma, 0 – 47 in/min

The torch is driven along one axis at a set speed and welds the straight corner seam of bent boxes, cabinets, trays and enclosures in mild steel, stainless steel and aluminum — with a TIG or a plasma source, instead of by hand. The operator places the part on the table with its corner against the external copper squaring guide and presses the lock button: a pneumatic cylinder presses it from inside against the copper guides, which hold the corner square and keep the heat of the part down. Start, and the machine welds the corner with the parameters of the recipe; release, and the next part goes in. The CWS 200 and CWS 300 take parts from 3-15/16 × 3-15/16" to 59-1/16 × 59-1/16" with corners 7-7/8" or 11-13/16" high; the WELCOR 400 X welds corners up to 15-3/4" high on parts to 39-3/8 × 39-3/8".

STANDARD FEATURES

- ✓ Welded steel base with guides that carry and move the torch unit
- ✓ Torch speed variable 0 – 47 in/min (0 – 20 mm/s), with acceleration and deceleration ramps
- ✓ Pneumatic clamping from inside against copper squaring guides that keep the heat of the part down
- ✓ Copper groups: outer unit 7-7/8" or 11-13/16" (200 or 300 mm) high; inner unit 1-15/16", 2-15/16", 3-15/16", 4-15/16", 5-7/8", 7-7/8", 11-13/16" (50, 75, 100, 125, 150, 200, 300 mm)
- ✓ CWS recipes: torch speed and its acceleration and deceleration ramps on the touch panel; welding current, start and end current ramps and gas pre-flow and post-flow on the welding power source
- ✓ WELCOR 400 X: numerical control storing 10 work parameters per job; signals a missing torch, gas, power or part
- ✓ Mild steel, stainless steel and aluminum; works with a TIG or a plasma welding power source
- ✓ CWS: work table 32-5/16 × 32-5/16" (820 × 820 mm); gap between the edges .008" (0.2 mm) at most
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

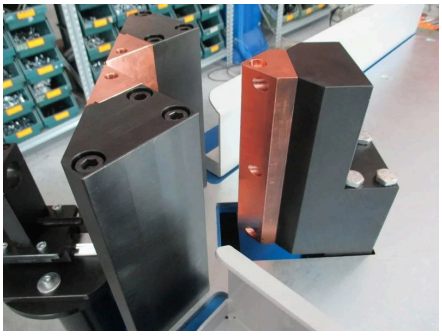
- + Inner copper groups with cooling — recommended to avoid dark marks on the parts
- + Special inner copper group designed for your part
- + Work table 39-3/8 × 39-3/8" (1,000 × 1,000 mm) (WELCOR 400 X)
- + Lorch T 250 TIG power source with cooling unit: T 250 DC for mild and stainless steel, T 250 AC/DC also for aluminum
- + Torch group with 9′ 10" (3 m) cable
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

DETAILS, COMPONENTS AND OPTIONS

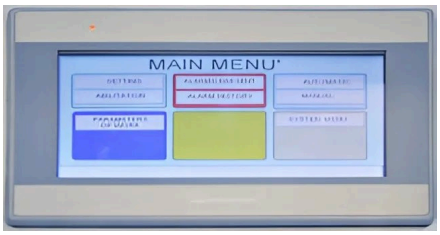
CWS 200 · CWS 300 · WELCOR 400 X 1-AXIS CORNER WELDERS, CORNERS TO 15-3/4" HIGH



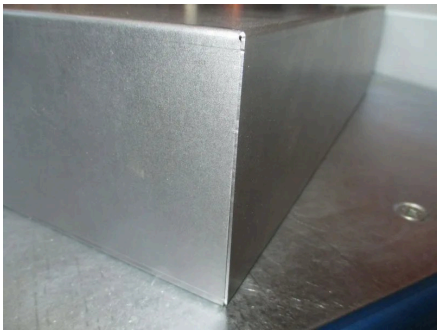
Copper groups that hold the corner and draw off the heat



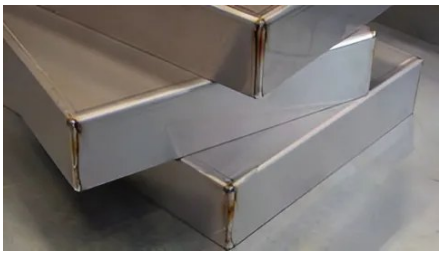
External copper squaring guide the corner rests against



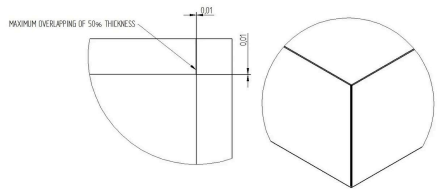
Touch panel of the CWS



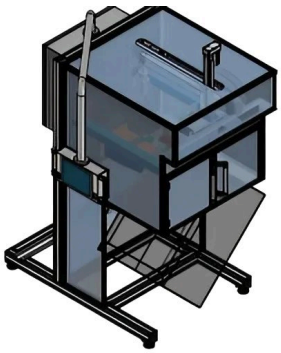
Bent corner before welding



Corner-welded boxes



How the corner is bent for welding: the edges .0004" (0.01 mm) apart and centred, or overlapping by half the sheet thickness at most



GECAM WELCOR 400 corner welder



Lorch T-series TIG power source — GECAM offers the T 250 with its corner welders

TECHNICAL SPECIFICATIONS

CWS 200 · CWS 300 · WELCOR 400 X 1-AXIS CORNER WELDERS, CORNERS TO 15-3/4" HIGH

1-AXIS CORNER WELDERS

CWS 200 · CWS 300 · WELCOR 400 X

Model	CWS 200	CWS 300	WELCOR 400 X
Torch axes	1 — X	1 — X	1 — X
Max. corner height	**7-7/8" (200 mm)**	**11-13/16" (300 mm)**	**15-3/4" (400 mm)**
Part size, min.	3-15/16 × 3-15/16" (100 × 100 mm)	3-15/16 × 3-15/16" (100 × 100 mm)	3-15/16 × 3-15/16" (100 × 100 mm)
Part size, max.	**59-1/16 × 59-1/16" (1,500 × 1,500 mm)**	**59-1/16 × 59-1/16" (1,500 × 1,500 mm)**	**39-3/8 × 39-3/8" (1,000 × 1,000 mm)**
Welding speed	**0 – 47 in/min (0 – 20 mm/s), variable**	**0 – 47 in/min (0 – 20 mm/s), variable**	**0 – 47 in/min (0 – 20 mm/s), variable**
Max. gap between the edges	.008" (0.2 mm)	.008" (0.2 mm)	—
Copper groups	Outer 7-7/8" (200 mm) · inner 1-15/16" – 11-13/16" (50 – 300 mm), 7 heights	Outer 11-13/16" (300 mm) · inner 1-15/16" – 11-13/16" (50 – 300 mm), 7 heights	Outer 7-7/8" or 11-13/16" (200 or 300 mm) · inner 1-15/16" – 11-13/16" (50 – 300 mm), 7 heights
Welding process	**TIG or plasma**	**TIG or plasma**	**TIG or plasma**
Materials	Mild steel, stainless steel, aluminum	Mild steel, stainless steel, aluminum	Mild steel, stainless steel, aluminum
Clamping	Pneumatic, from inside, against copper squaring guides	Pneumatic, from inside, against copper squaring guides	Pneumatic, from inside, against copper squaring guides
Control	Touch panel, parameters stored as recipes	Touch panel, parameters stored as recipes	Numerical control, 10 work parameters stored per job
Work table	32-5/16 × 32-5/16" (820 × 820 mm)	32-5/16 × 32-5/16" (820 × 820 mm)	—

Corner height is the longest corner seam the torch welds, on each axis. The welding power source is not included: GECAM offers the Lorch T 250 TIG source, and the machines also work with a plasma source. Bend the corner so the two edges meet on it — .0004" (0.01 mm) apart and centred on the corner — or, where the bend does not allow it, so they overlap by half the sheet thickness at most. Metric values in brackets are the manufacturer's figures.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Italy by GECAM — 100 % Italian
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
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  in-stock machines only
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- ✓ Installation and training available

CWS 300 XY · WELCOR 400 XY · 2 MODELS

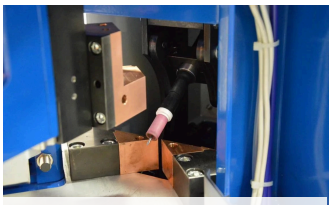
CWS 300 XY · WELCOR 400 XY 2-AXIS CORNER WELDERS, PARTS TO 59 × 59″



MODEL 2-AXIS · GECAM CWS 300 XY TWO-AXIS CORNER WELDING MACHINE



Box corner clamped against the copper groups, TIG torch at the seam



Torch and copper corner group of the CWS 300 XY

Two axes, X and Y · CWS 300 XY parts to 59″ square · WELCOR 400 XY corners 15-3/4″ on X, 4-5/16″ on Y

Here the torch is driven on two axes, X and Y. The CWS 300 XY takes parts from 3-15/16 × 3-15/16″ to 59-1/16 × 59-1/16″ with corners 7-7/8″ to 11-13/16″ high; its touch panel shows the X and Y positions, counts the parts and calls up the recipe. The WELCOR 400 XY welds corners up to 15-3/4″ on X and 4-5/16″ on Y — a seam that runs in two directions, in one cycle — on parts to 39-3/8 × 39-3/8″. The rest works as on the one-axis machines: the part is clamped pneumatically from inside against copper squaring guides, the torch welds at 0 to 47 in/min with a TIG or a plasma source, and the parameters of each part are stored.

STANDARD FEATURES

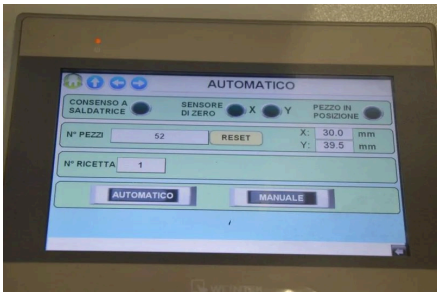
- ✓ Torch driven on two axes, X and Y; on the WELCOR 400 XY corners up to 15-3/4″ (400 mm) on X and 4-5/16″ (110 mm) on Y
- ✓ Welding speed variable 0 – 47 in/min (0 – 20 mm/s), with acceleration and deceleration ramps
- ✓ Welded steel base with guides that carry and move the torch unit
- ✓ Pneumatic clamping from inside against copper squaring guides that keep the heat of the part down
- ✓ Copper groups: outer unit 7-7/8″ or 11-13/16″ (200 or 300 mm) high; inner unit 1-15/16″, 2-15/16″, 3-15/16″, 4-15/16″, 5-7/8″, 7-7/8″, 11-13/16″ (50, 75, 100, 125, 150, 200, 300 mm)
- ✓ CWS 300 XY: touch panel with the X and Y positions, part counter and recipe number; welding current, current ramps and gas flow set on the welding power source
- ✓ WELCOR 400 XY: numerical control storing 10 work parameters per job; signals a missing torch, gas, power or part
- ✓ Mild steel, stainless steel and aluminum; TIG or plasma
- ✓ CWS 300 XY: work table 32-5/16 × 32-5/16″ (820 × 820 mm)
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

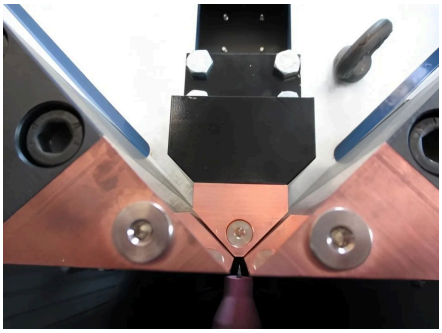
- + Work table 39-3/8 × 39-3/8″ (1,000 × 1,000 mm)
- + Special inner copper group designed for your part
- + Inner copper group with cooling
- + Lorch T 250 TIG power source with cooling unit: T 250 DC for mild and stainless steel, T 250 AC/DC also for aluminum
- + Torch group with 9′ 10″ (3 m) cable
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

DETAILS, COMPONENTS AND OPTIONS

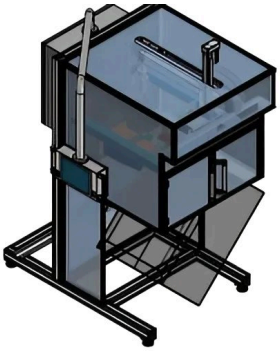
CWS 300 XY · WELCOR 400 XY 2-AXIS CORNER WELDERS, PARTS TO 59 × 59″



Touch panel of the CWS 300 XY: X and Y positions, part counter, recipe number



Inner copper group pressing the corner from inside, torch below



GECAM WELCOR 400 corner welder

TECHNICAL SPECIFICATIONS

CWS 300 XY · WELCOR 400 XY 2-AXIS CORNER WELDERS, PARTS TO 59 × 59″

2-AXIS CORNER WELDERS

CWS 300 XY · WELCOR 400 XY

Model	CWS 300 XY	WELCOR 400 XY
Torch axes	**2 — X, Y**	**2 — X, Y**
Max. corner height	**7-7/8″ – 11-13/16″ (200 – 300 mm)**	**X 15-3/4″ (400 mm) · Y 4-5/16″ (110 mm)**
Part size, min.	3-15/16 × 3-15/16″ (100 × 100 mm)	3-15/16 × 3-15/16″ (100 × 100 mm)
Part size, max.	**59-1/16 × 59-1/16″ (1,500 × 1,500 mm)**	**39-3/8 × 39-3/8″ (1,000 × 1,000 mm)**
Welding speed	**0 – 47 in/min (0 – 20 mm/s), variable**	**0 – 47 in/min (0 – 20 mm/s), variable**
Copper groups	Outer 7-7/8″ or 11-13/16″ (200 or 300 mm) · inner 1-15/16″ – 11-13/16″ (50 – 300 mm), 7 heights	Outer 7-7/8″ or 11-13/16″ (200 or 300 mm) · inner 1-15/16″ – 11-13/16″ (50 – 300 mm), 7 heights
Welding process	**TIG or plasma**	**TIG or plasma**
Materials	Mild steel, stainless steel, aluminum	Mild steel, stainless steel, aluminum
Clamping	Pneumatic, from inside, against copper squaring guides	Pneumatic, from inside, against copper squaring guides
Control	Touch panel: X and Y positions, part counter, recipes	Numerical control, 10 work parameters stored per job
Work table	32-5/16 × 32-5/16″ (820 × 820 mm)	—
Work table, to order	39-3/8 × 39-3/8″ (1,000 × 1,000 mm)	39-3/8 × 39-3/8″ (1,000 × 1,000 mm)

Corner height is the longest corner seam the torch welds, on each axis. The welding power source is not included: GECAM offers the Lorch T 250 TIG source, and the machines also work with a plasma source. Bend the corner so the two edges meet on it — .0004″ (0.01 mm) apart and centred on the corner — or, where the bend does not allow it, so they overlap by half the sheet thickness at most. Metric values in brackets are the manufacturer's figures.

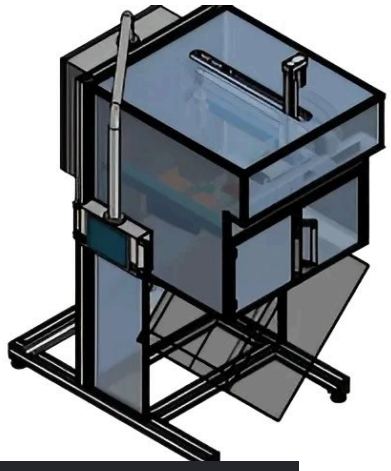
Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

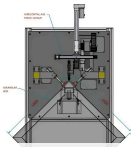
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- ✓ 1-year warranty
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  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

WELCOR 400 XYZ · 3 AXES

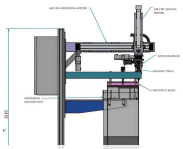
WELCOR 400 XYZ 3-AXIS CORNER WELDER FOR SEAMS IN THREE DIRECTIONS, PARTS TO 39-3/8 × 39-3/8″



MODEL 3-AXIS · GECAM WELCOR 400 CORNER WELDER



WELCOR 400 front view with the largest box, 39-3/8 × 39-3/8″ (1,000 × 1,000 mm), corner up — machine 43-5/16″ (1,100 mm) wide



WELCOR 400 side view: torch axes, welding torch and pneumatic block — machine 87″ (2,210 mm) high, 61-3/4″ (1,569 mm) deep

Three axes, X, Y and Z · corners 15-3/4″ X, 4-5/16″ Y, 1-15/16″ Z · parts to 39-3/8″ square

With three driven axes — X, Y and Z — the torch follows a corner seam that runs in three directions and welds it in one cycle: up to 15-3/4″ on X, 4-5/16″ on Y and 1-15/16″ on Z, on parts from 3-15/16 × 3-15/16″ to 39-3/8 × 39-3/8″. The numerical control stores the work parameters of each job — torch speed with its acceleration and deceleration ramps, the welding speed set on the machine, interruptions of the weld, welding power and welding speed ramps up and down, power supply, shielding gas and cooling of the part during welding — and the machine signals a missing torch, welding gas, gas protection, welding power, electrical connection or part. The part is clamped pneumatically from inside against copper squaring guides; TIG or plasma, 0 to 47 in/min.

STANDARD FEATURES

- ✓ Torch driven on three axes: corners up to 15-3/4″ (400 mm) on X, 4-5/16″ (110 mm) on Y and 1-15/16″ (50 mm) on Z
- ✓ Welding speed variable 0 – 47 in/min (0 – 20 mm/s), adjusted directly on the machine
- ✓ Ten work parameters stored per job: torch speed, acceleration and deceleration ramps, welding speed, weld interruption, welding power ramps up and down, welding speed ramps up and down, power supply on/off, shielding gas on/off, cooling of the part during welding
- ✓ Signals a missing torch, welding gas, gas protection, welding power, electrical connection or part
- ✓ Welded steel base with guides that carry and move the torch unit
- ✓ Pneumatic clamping from inside against copper squaring guides; outer copper unit 7-7/8″ or 11-13/16″ (200 or 300 mm), inner unit 1-15/16″, 2-15/16″, 3-15/16″, 4-15/16″, 5-7/8″, 7-7/8″, 11-13/16″ (50, 75, 100, 125, 150, 200, 300 mm)
- ✓ Mild steel, stainless steel and aluminum; TIG or plasma
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Work table 39-3/8 × 39-3/8″ (1,000 × 1,000 mm)
- + Special inner copper group designed to the drawing of your part
- + Cooling unit for the inner copper group
- + External cooling unit
- + Lorch T 250 TIG power source: T 250 DC for mild and stainless steel, T 250 AC/DC also for aluminum
- + Torch group with 9′ 10″ (3 m) cable
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS

WELCOR 400 XYZ 3-AXIS CORNER WELDER FOR SEAMS IN THREE DIRECTIONS, PARTS TO 39-3/8 × 39-3/8″

3-AXIS CORNER WELDER

WELCOR 400 XYZ

Model	WELCOR 400 XYZ
Torch axes	**3 — X, Y, Z**
Max. corner height, X axis	**15-3/4″ (400 mm)**
Max. corner height, Y axis	**4-5/16″ (110 mm)**
Max. corner height, Z axis	**1-15/16″ (50 mm)**
Part size, min.	3-15/16 × 3-15/16″ (100 × 100 mm)
Part size, max.	**39-3/8 × 39-3/8″ (1,000 × 1,000 mm)**
Welding speed	**0 – 47 in/min (0 – 20 mm/s), variable**
Copper groups	Outer 7-7/8″ or 11-13/16″ (200 or 300 mm) · inner 1-15/16″ – 11-13/16″ (50 – 300 mm), 7 heights
Welding process	**TIG or plasma**
Materials	Mild steel, stainless steel, aluminum
Clamping	Pneumatic, from inside, against copper squaring guides
Control	Numerical control, 10 work parameters stored per job
Signals	Missing torch, welding gas, gas protection, welding power, electrical connection or part
Work table, to order	39-3/8 × 39-3/8″ (1,000 × 1,000 mm)

Corner height is the longest corner seam the torch welds, on each axis. The welding power source is not included: GECAM offers the Lorch T 250 TIG source, and the machines also work with a plasma source. Bend the corner so the two edges meet on it — .0004″ (0.01 mm) apart and centred on the corner — or, where the bend does not allow it, so they overlap by half the sheet thickness at most. Metric values in brackets are the manufacturer's figures.

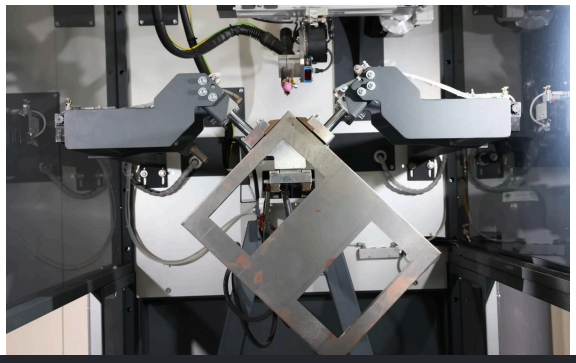
Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

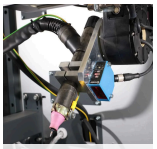
- ✓ Made in Italy by GECAM — 100 % Italian
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

WELCOR 400 XYZ-Φ · 4 AXES

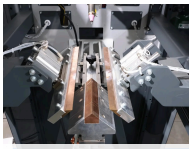
WELCOR 400 XYZ-Φ 4-AXIS CORNER WELDER WITH LASER SEAM READING, PARTS TO 39-3/8 × 39-3/8″



MODEL 4-AXIS · GECAM WELCOR 400 4-AXIS WELDING THE CORNER OF A SQUARE FRAME



Laser sensor beside the TIG torch on the rotary actuator



Pneumatic clamps and copper pads holding the corner

Four axes, X, Y, Z and torch rotation · laser seam reading · parts to 39-3/8″ square · 0 – 47 in/min

The three axes of the WELCOR 400 XYZ and a fourth: the torch is mounted on a rotary actuator that sets its angle in each phase of the weld, from vertical to about 120° as the job needs. Before it welds, the torch runs along the weld line while a laser sensor beside it checks that the edges are straight and how long they are; the machine memorizes the path and then welds it — a self-learning system that finds the corners by itself. Programs are saved for repeat work, the automatic welding program makes the machine easy to run even for operators who are not welders, and the color touch screen draws the welding process to help pick the right parameters. Pneumatic clamping positions the flaps; modular copper pads, matched to the size of the box, carry off the heat and limit burning of the part. Parts from 3-15/16 × 3-15/16″ to 39-3/8 × 39-3/8″, the smallest box 1-15/16″ high; torch path up to 15-3/4 × 1-7/8 × 4-7/16″.

STANDARD FEATURES

- ✓ Four axes: X, Y, Z and torch rotation Φ — travel 15-3/4″ (400 mm) on X, 4-5/16″ (110 mm) on Y, 7-7/8″ (200 mm) on Z
- ✓ Torch on a rotary actuator, 0 – 90°: its angle is set for each phase of the weld, from vertical to about 120° to the job
- ✓ Laser sensor beside the torch: the torch runs the weld line first, the laser checks the straightness and length of the edges, the path is memorized, then welded
- ✓ Color touch screen with a drawing of the welding process; programs saved for repeat parts; automatic welding program for operators who are not welders
- ✓ Ten work parameters per job — torch speed and its ramps, welding speed, weld interruption, welding power and speed ramps, power supply, shielding gas, cooling of the part
- ✓ Signals a missing torch, welding gas, gas protection, welding power, electrical connection or part
- ✓ Pneumatic clamping that positions the flaps; modular copper groups — aluminum support and copper unit — 3/8″, 13/16″, 1″, 1-3/16″, 1-15/16″, 3-15/16″, 7-7/8″ (10, 20, 25, 30, 50, 100, 200 mm) thick, matched to the box
- ✓ Welding speed variable 0 – 47 in/min (0 – 20 mm/s)
- ✓ Mild steel, stainless steel and aluminum; TIG or plasma
- ✓ Overall dimensions 59 × 47 × 98″ (1,500 × 1,200 × 2,500 mm)
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

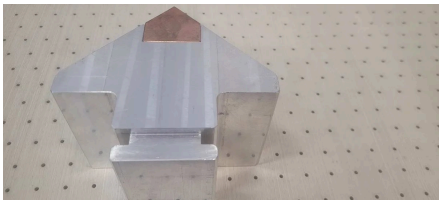
- + Lorch T 250 TIG power source: T 250 DC for mild and stainless steel, T 250 AC/DC also for aluminum
- + External cooling unit
- + Torch group with 9′ 10″ (3 m) cable
- + Special inner copper group designed to the drawing of your part
- + Cooling unit for the inner copper group
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

DETAILS, COMPONENTS AND OPTIONS

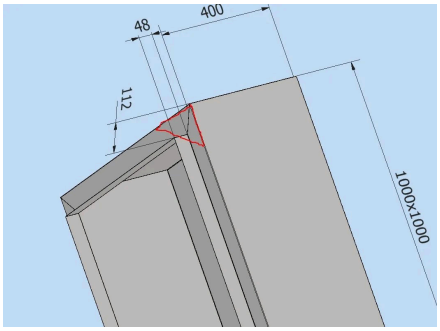
WELCOR 400 XYZ-Φ 4-AXIS CORNER WELDER WITH LASER SEAM READING, PARTS TO 39-3/8 × 39-3/8”



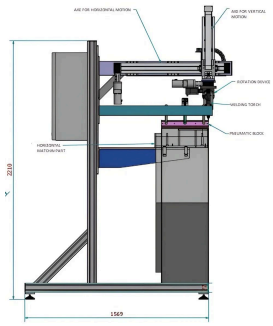
Control panel with color touch screen and signal tower



Copper group: aluminum support and copper unit, 3/8” to 7-7/8” (10 – 200 mm) thick



Largest torch path, 15-3/4 × 1-7/8 × 4-7/16” (400 × 48 × 112 mm), on a box up to 39-3/8 × 39-3/8” (1,000 × 1,000 mm)



WELCOR 400 side view: torch axes, welding torch and pneumatic block — machine 87” (2,210 mm) high, 61-3/4” (1,569 mm) deep

TECHNICAL SPECIFICATIONS

WELCOR 400 XYZ-Φ 4-AXIS CORNER WELDER WITH LASER SEAM READING, PARTS TO 39-3/8 × 39-3/8″

4-AXIS CORNER WELDER

WELCOR 400 XYZ-Φ

Model	WELCOR 400 XYZ-Φ
Torch axes	**4 — X, Y, Z and torch rotation Φ**
Max. corner height, X axis	**15-3/4″ (400 mm)**
Max. corner height, Y axis	**4-5/16″ (110 mm)**
Max. corner height, Z axis	**1-15/16″ (50 mm)**
Axis travel	X 15-3/4″ · Y 4-5/16″ · Z 7-7/8″ (400 · 110 · 200 mm)
Working range of an axis	3/8″ – 15-3/4″ (10 – 400 mm)
Torch rotation	**0 – 90° · from vertical to about 120° to the job**
Part size, min.	3-15/16 × 3-15/16″ (100 × 100 mm) · smallest box 1-15/16″ (50 mm) high
Part size, max.	**39-3/8 × 39-3/8″ (1,000 × 1,000 mm)**
Max. torch path	15-3/4 × 1-7/8 × 4-7/16″ (400 × 48 × 112 mm)
Welding speed	**0 – 47 in/min (0 – 20 mm/s), variable**
Seam reading	**Laser sensor beside the torch: edges checked, path memorized, then welded**
Control	Color touch screen with a drawing of the welding process · programs saved · automatic welding program
Work parameters	10 per job
Signals	Missing torch, welding gas, gas protection, welding power, electrical connection or part
Clamping	Pneumatic, positioning the flaps
Copper groups	Modular: aluminum support + copper unit, 3/8″ – 7-7/8″ (10 – 200 mm) thick
Welding process	**TIG or plasma**
Materials	Mild steel, stainless steel, aluminum
Dimensions L × W × H	59 × 47 × 98″ (1,500 × 1,200 × 2,500 mm)

Corner height is the longest corner seam the torch welds, on each axis. The welding power source is not included: GECAM offers the Lorch T 250 TIG source, and the machines also work with a plasma source. Bend the corner so the two edges meet on it — .0004″ (0.01 mm) apart and centred on the corner — or, where the bend does not allow it, so they overlap by half the sheet thickness at most. Metric values in brackets are the manufacturer's figures. The WELCOR 4-axis brochure gives 5-7/8 × 5-7/8″ (150 × 150 mm) as the smallest workable part, the technical specification 3-15/16 × 3-15/16″ (100 × 100 mm) — send us the part to confirm.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

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Specifications subject to change without notice. Imperial values are converted from manufacturer metric data and rounded: thin sheet in gauge (ga), plate and capacities in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment. 7-day money-back guarantee on in-stock machines only; financing available.